

## 🧩 RS003 Week 13 — English Worksheet

**Topic:** A Second Function · **Course:** Pygame Turret · **Time:** about 45 minutes

This week you split your game into **several small functions** — one for the ground, one for the turret, one for a target, one for the aim marker. The functions work together by sharing the gun's X position, and the marker changes colour when the aim lines up with the target.

💬 Words to know: **function**, **call**, **share a value**, **target**, **marker**

### 1 · Predict 🧠

Read each piece of code. Before you run it, write what you think will happen.

```
def draw_ground():
    pygame.draw.rect(screen, GREEN, (0, HEIGHT - 50, WIDTH, 50))

def draw_turret(gun_x):
    ...

draw_ground()
draw_turret(CENTER_X)
```

**draw\_ground** takes no input but **draw\_turret** takes a **gun\_x**. Why does only one of them need a value?

|  |
|--|
|  |
|  |
|  |
|  |

```
def draw_marker(gun_x):
    distance = abs(gun_x - TARGET_X)
    color = YELLOW if distance < 20 else RED
    pygame.draw.circle(screen, color, (gun_x, HEIGHT - 200), 8)
```

`abs(gun_x - TARGET_X)` is how far the aim is from the target. When does the marker turn yellow?

|  |
|--|
|  |
|  |
|  |
|  |

```
draw_ground()  
draw_target()  
draw_turret(gun_x)  
draw_marker(gun_x)
```

Four functions are called in a row. Which one is drawn last, and what does that mean for what sits in front?

|  |
|--|
|  |
|  |
|  |
|  |

---

## 2 · Run It

 **Type the example, run it, and line up the target**

```

import pygame
pygame.init()

WIDTH, HEIGHT = 800, 600
CENTER_X = WIDTH // 2
screen = pygame.display.set_mode((WIDTH, HEIGHT))

SKY = (135, 206, 235)
GREEN = (0, 100, 0)
GRAY = (80, 80, 80)
RED = (200, 0, 0)
YELLOW = (255, 255, 0)
BLACK = (0, 0, 0)

TARGET_X = 600
TARGET_Y = HEIGHT - 200

def draw_ground():
    pygame.draw.rect(screen, GREEN, (0, HEIGHT - 50, WIDTH, 50))

def draw_turret(gun_x):
    pygame.draw.rect(screen, GREEN, (CENTER_X - 30, HEIGHT - 90, 60, 40))
    pygame.draw.line(screen, GRAY, (CENTER_X, HEIGHT - 90), (gun_x, HEIGHT - 200),
10)

def draw_target():
    pygame.draw.circle(screen, BLACK, (TARGET_X, TARGET_Y), 25)
    pygame.draw.circle(screen, RED, (TARGET_X, TARGET_Y), 18)
    pygame.draw.circle(screen, YELLOW, (TARGET_X, TARGET_Y), 8)

def draw_marker(gun_x):
    distance = abs(gun_x - TARGET_X)
    color = YELLOW if distance < 20 else RED
    pygame.draw.circle(screen, color, (gun_x, HEIGHT - 200), 8)

gun_x = CENTER_X
SPEED = 5
MIN_X, MAX_X = 100, WIDTH - 100

clock = pygame.time.Clock()
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

```

```
keys = pygame.key.get_pressed()
if keys[pygame.K_LEFT] and gun_x > MIN_X:
    gun_x -= SPEED
if keys[pygame.K_RIGHT] and gun_x < MAX_X:
    gun_x += SPEED

screen.fill(SKY)
draw_ground()
draw_target()
draw_turret(gun_x)
draw_marker(gun_x)
pygame.display.flip()
clock.tick(60)
```

```
pygame.quit()
```

**Aim slowly toward the target. What happens to the marker colour as you get close?**

|  |
|--|
|  |
|  |
|  |
|  |

### 3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it, then explain why the original was wrong.

**Bug A** — The marker should follow the gun, but it is stuck in the centre and never moves.

```
def draw_marker():  
    pygame.draw.circle(screen, RED, (CENTER_X, HEIGHT - 200), 8)
```

**Hint:** the marker needs the gun's position passed in.

**Write the fixed code (the whole function):**

|  |
|--|
|  |
|  |
|  |
|  |

**Why was it wrong? Why does your fix work?**

|  |
|--|
|  |
|  |
|  |
|  |

**Bug B** — The target should sit **behind** the marker, but right now the target circle covers the marker so you cannot see when you are aiming on it.

```
draw_ground()  
draw_turret(gun_x)  
draw_marker(gun_x)  
draw_target()
```

**Hint:** whatever is drawn last sits on top.

**Write the fixed code (calls in the right order):**

|  |
|--|
|  |
|  |
|  |
|  |

**Why was it wrong? Why does your fix work?**

|  |
|--|
|  |
|  |
|  |
|  |

**Bug C** — The marker is supposed to turn yellow only when the aim is **close** to the target, but it is yellow everywhere.

```
distance = gun_x - TARGET_X
color = YELLOW if distance < 20 else RED
```

**Hint:** without `abs`, a big negative distance is also "less than 20".

**Write the fixed code:**

|  |
|--|
|  |
|  |
|  |
|  |

**Why was it wrong? Why does your fix work?**

|  |
|--|
|  |
|  |
|  |
|  |

---

## 4 · Modify It

### Move the target somewhere new

Change `TARGET_X` so the target sits in a different spot, then aim at it.

**Write the new TARGET\_X you chose:**

|  |
|--|
|  |
|  |
|  |
|  |

### Add a fourth function of your own

Write one more small function — for example `draw_score()` or `draw_clouds()` — and call it in the loop.

**Write your function name and what it draws:**

|  |
|--|
|  |
|  |
|  |
|  |

**Hint:** a function is just a name for a group of drawing lines. If it needs a changing value (like the gun position), pass it in as a parameter.



## 5 · Make It 📷

## Build a game split into functions

Build a program where:

1. the ground, turret, target, and marker are each their own function,
2. the functions share the gun's position,
3. the marker changes colour when you aim on the target.

Send a **photo or video** of your aimed-on target, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I split my code into functions for ...

The functions share the gun position by ...

The marker turns yellow when ...

I used `abs(...)` to ...

One tricky moment was when ...

If I had more time, I would ...

## 6 · Record Your Walkthrough 🎥

Take a video while you aim onto the target. Talk through it like you are teaching someone. Try to use these words: **function**, **call**, **share**, **target**, **marker**.

1. Show the marker turning yellow on the target.
2. Read your function names out loud and say what each one draws.
3. Explain how the gun position is shared between functions.
4. Point at the line that decides the marker colour.

**Write what you will say in your video.** Plan it here before you record.

|  |
|--|
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |
|  |

**Submit** ✓

Send this worksheet + your photo or video to teacher on KakaoTalk.